Based on the logic we programmed into your BiomechanicsAnalyzer class and the coaching advice dictionary, a **"Good Player"** is mathematically defined as someone who adheres to the **"Textbook Orthodoxy"** of cricket.

In your specific AI system, a "Good Player" (one who scores **85-100/100**) meets the following 5 strict biomechanical criteria at the moment of impact:

1. The "High Elbow" Extension (Power Lever)

- **The Criteria:** The angle formed by the Shoulder-Elbow-Wrist must be between **120° and 140°**.
- **Why?**
  - **Physics:** A straight(er) arm creates a longer lever. A longer lever creates more torque and speed at the end of the bat (Impact point).
  - **Error:** If the angle is <110° ("Chicken Wing"), the player is cramping themselves and relying only on forearm strength, not leverage.

2. The "Still Head" (Visual Stability)

- **The Criteria:** The vertical movement of the Nose coordinate across the entire shot sequence must be **less than 10cm**.
- **Why?**
  - **Perception:** If the head drops or lifts, the eyes lose the trajectory of the ball. A still head ensures the batter judges the length of the ball correctly.
  - **Balance:** The head is the heaviest part of the body. If it falls over (drift > 15cm), the body weight collapses, and the shot loses power.

3. The "Anchored" Base (Kinetic Chain)

- **The Criteria:** The Back Foot (Ankle Y-coordinate) must not lift more than **5cm** vertically before/during impact.
- **Why?**
  - **Stability:** You cannot generate force from an unstable base ("You can't fire a cannon from a canoe").
  - **Weight Transfer:** The back foot must act as an anchor to push weight into the front foot. If it lifts early (>10cm), the batter is "jumping" at the ball rather than driving through it.

4. The "Coiled" Hips (Torque Generation)

- **The Criteria:** The angle difference of the hip line between Stance and Impact must be:
    - **> 30°** for Front Foot shots (Drive/Cut)
    - **> 60°** for Cross Bat shots (Pull/Sweep)
- **Why?**
    - **Core Power:** The big muscles are in the legs and core. Rotating the hips "uncoils" the body, transferring massive energy into the bat.
    - **Error:** Minimal rotation (<15° error margin) means the player is just using their arms ("Handsy" shot), resulting in weak strokes.

5. The "High Finish" (Acceleration)

- **The Criteria:** At the final frame, the **Wrist Y-coordinate must be higher** (numerically smaller in pixels) than the Shoulder Y-coordinate.
- **Why?**
    - **Commitment:** To finish high, the bat must accelerate *through* the ball.
    - **Deceleration:** If hands finish low (below shoulder), it means the batter started slowing down the bat *before* they even hit the ball. A high finish guarantees acceleration through the contact zone.

Summary Table for Your Defense Presentation

You can copy this table into your slides to explain exactly how your AI grades a player:

Biometric Feature	**Good Player (Pro)**	**Average Player**	**Poor Form (Beginner)**	**Why it matters?**
Elbow Extension	**120° - 140°**	110° - 120°	< 110° (Bent)	Maximizes lever length & bat speed.
Head Stability	**< 10 cm drift**	10 - 15 cm	> 15 cm drift	Ensures eye-line stability & balance.

Biometric Feature	Good Player (Pro)	Average Player	Poor Form (Beginner)	Why it matters?
Back Foot Lift	**< 5 cm (Grounded)**	5 - 10 cm	> 10 cm (Airborne)	Provides a solid base for power transfer.
Hip Rotation	**> 30° (Drive) / > 60° (Pull)**	15° - 30°	< 15° (Static)	Generates torque from the core.
Follow Through	**Hands > Shoulders**	Hands ≈ Shoulders	Hands < Shoulders	Ensures acceleration through impact.

In short: Your AI defines a "Good Player" as someone who maximizes leverage, maintains balance, generates core power, and accelerates through the ball.